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Curating Hybrid-Style University Experiences: Framework Development for Student Social Connectedness Using Interaction Design and Placemaking

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Abstract: Communication technologies offer new opportunities for building and maintaining social connections at university. However, these technologies need to be better understood for their influence on student experiences in hybrid-style classrooms and guided more effectively. This study identified qualities of social connection to understand how students built/maintained social connection. The study was conducted over two autumn semesters in three hybrid-style classes in Switzerland and Norway. Qualitative methods included observations, interviews, and auto-photography. Approximately 150 h of observational data were gained, including auto-photography of 17 students' physical online learning spaces, and four interviews with class lecturers. The findings indicated that informal/private spaces, crucial for building/maintaining social connection, were hard to access in hybrid-style learning environments. Communication cultures and social structures offered limited support for accessing these spaces for building/maintaining social connection. Addressing this issue required a processual effort across user-levels to support social connection within a hybrid-style university culture. This effort involved opening access to informal/private spaces by applying the concept of affective atmospheres within placemaking and interaction-design theories. A dimensions of affect framework is developed to help curate learning environments towards social connection. For university campus development, addressing these people–place relations with communication technology could foster technology more effectively to support connection.

Keywords: learning environment; hybrid-style university; student social connectedness; university experience; framework development; affective atmospheres

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Citation: Wheele, T.; Lindkvist, C.; Haugen, T.; Weber, C.; Windlinger, L. Curating Hybrid-Style University Experiences: Framework Development for Student Social Connectedness Using Interaction Design and Placemaking. *Educ. Sci.* **2024**, *14*, 1004. https://doi.org/10.3390/educsci14091004

Academic Editor: Xinqiao Liu

Received: 13 June 2024
Revised: 4 September 2024
Accepted: 6 September 2024
Published: 12 September 2024

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1. Introduction

Better guidelines are required to support facility managers and organizational disciplines in making decisions on university campus development in the digitalized age. The aim of this research is to provide a compass for lecturers, program leaders, spatial planners and managers, and policymakers to use when making decisions about managing or designing for the student experience in a digitally oriented university course. This requires new ways of thinking about the management of space to attend to people–place-technology relations. To target these relations, theories from placemaking and interaction design are addressed and united through the concept of affective atmospheres. Affective atmospheres offer a means of bringing recognition to the 'what' of student experiences with social connection, helping to understand qualities that influence the experience. Affective atmospheres, as a concept, help identify atmospheres as imagined, constructed, and conceived as affective for curating specific experiences [1]. Following the results from Wheele et al. [2], four

dimensions of affect in the learning environment are identified, namely materiality, adaptability, temporality, and cultural/sociality. This process of understanding atmospheres suggests that a digitally oriented university learning environment can be curated and guided to support student experiences with social connectedness. This is explored through placemaking and interaction-design theories in a proposed three-step principle for curating an experience taken from the 'why, what, how' principle and the 'inspiration, ideation, implementation' principle [3,4]. The aim of a framework is to help navigate the qualities that constitute student social connectedness in the digitally oriented learning environment and guide university campus development towards managing student experiences with a social connection in the future university. This is valuable for campus planners and facility management and services for university learning environments.

The COVID-19 pandemic has had an impact on education, shifting it towards digitalization. The pandemic has, in many ways, passed, yet communication technologies continue to be desired and integrated into the learning environment. Although there were slow progressions towards the digitalization of education pre-pandemic, the pandemic thrust universities into accepting digitalization as an emergency response [5,6]. COVID-19 brought rapid and irregular adoption of communication technologies at university with little guidance on the best ways to adopt this for teaching and learning. Unplanned emergency responses integrated communication technology at university on a mass scale. While they demonstrated the potential of technology in the context of teaching and learning, they also demonstrated its limits [7]. In the learning environment, communication technologies have proven to bring new ways to connect and disconnect through more personalized and flexible learning experiences [8,9]. Post-pandemic, digitalization continues to be integrated to various extents across education, with the highest rates of adoption in higher education [10]. A hybrid-style approach to learning has become popular amongst university students because it adopts a bit of both worlds, being both online (virtual) and in person (physical). With technology, a hybrid style of learning can encompass a flexible and personalized approach while still maintaining an aspect of the physical university campus. This also accounts for digital technologies as inherently entwined in our everyday lives, setting students up for the future, as hybrid workplaces become a growing expectation [11]. Technologies in this way have opened the door for university campus development by bringing new ways to connect and disconnect, and these opportunities have brought new experiences for students at university [7].

Communication technologies are continually being developed to satisfy gaps in the market (e.g., poor-quality video setups aim to be alleviated by companies like Owl Labs) and develop a more successful learning experience, but this does not address the whole picture. Being online shifts how connections are made. It is proposed that building and maintaining social connections require private and informal spaces, as found by Wheele et al. [12]. In this, private and informal space might not be prototypically represented, since communication technology brings an opening of liminal space, which is little understood and can be both 'closed' and 'open' in a blurred representation of space [13]. Thus, while some online communication platforms offer spaces of this kind (e.g., WhatsApp or social media platforms like Instagram or Facebook), they are highly difficult to manage by university management in the university system, and this should also not be the aim. Yet, with online management platforms replacing campuses in some cases [10], this level of technology integration brings concerns for education and society as they lose control when trying to avoid issues that are emerging, like a greater risk of distraction and a lack of human contact for students [10]. This is a concern because poor social connection at university has a negative impact on learning and social development, including aspects such as engagement and identity development [14]. While these communication technologies are bringing new opportunities for students to have greater freedom and a lack of need for a physical university place, the physical university place remains important if universities are to still have access when supporting the student experience. So much so that Allen and McLaren [15] identify the physical university campus as impossible to replicate digitally.

Thus, it is important to recognize the role of the physical campus alongside the limits of managing communication technology to understand how to utilize technology effectively. The content of this article is important to help develop the theory on student experiences of hybrid-style learning environments. Further, by supporting framework development, it has the scope to transfer findings and integrate them into practice.

The article first outlines the two problem statements and questions of this study that center on (1) identifying student social connection and (2) curating student social connection. This section is followed by an outline of the theoretical perspectives of interaction design and placemaking with the concept of affective atmospheres. This section is followed by an overview of the methodology to outline the qualitative approaches using observations, auto-photography, and interviews. The key findings of the study are outlined, followed by a discussion provided in two parts to address the two research problems and questions through (1) accessing spaces of social connection and (2) a framework of experience. Limitations of the research are identified to include the small sample size and the inherent limitations of accessing observational data of students behind the online screen. Finally, the conclusion section summarizes the study and its contribution to better understanding communication technologies for student experiences in hybrid-style classrooms and how to guide them more effectively.

1.1. Problem Statement

1.1.1. Step 1: Identifying Student Social Connection

The problem of identifying student social connection is addressed in the first aim of the article through research question RQ1, namely "How is student social connectedness built and maintained in spaces of the digitally-orientated university?". Communication technologies have the potential to support and increase connections between people and people and people and places, yet they are largely misunderstood in how they should be managed to support the student experience [5]. Technologies bring new opportunities for social connection that can expand learning and social development [6,16], where an education without technology is deemed unlikely to be as relevant in society today [10]. As the global education monitoring report by UNESCO [10] stresses, education systems must find ways to use technology as a tool to support but not replace human interaction. While it is evident that combining online and offline space would be optimal for learning at university, technologies are not fully understood in terms of how they aid or hinder students [5,17]. This leads to technologies not always being used effectively or to their full potential by both students and university staff to support connection in the learning environment. These challenges with technology and social connection are significant for education because the university has the role of being both an academic and social space, where students develop social skills alongside academic credentials [10]. Without the constant physical tie to being on or around campus, students must renegotiate how they build and maintain social connections through technology. New ways of conducting research are needed to support and manage the impact of technology on social concerns in the spaces of the university. This is because technology changes the way people see and experience the world, which also requires changes to the way research should be conducted to accommodate for technology-related social concerns that are less tangible or straightforward to address in management strategies. Thus, while UNESCO [10] recognizes three conditions that must be met for technology in education to be used to its full potential, namely access to technology, governance regulation, and teacher preparation, implementing this requires an understanding of people–place–technology relations.

Identifying the qualities that influence people–place–technology relations requires innovative research approaches that transcend disciplinary boundaries. This is because technology opens the potential for 'in-between' space to emerge as a blurred and liminal experience in space [13]. This 'in-between' space supports or weakens student social connection in different ways, as technology allows for the disembodying of people in space. Individuals become physically detached from virtual space, despite continuing to

create or be influenced by experiences of virtual place through practices, meanings, and identities, bringing a sense of placelessness, being 'without place', or having a 'Second Life' [18–20]. These are highly intangible alterations to our experiences of people–people and people–place relations that may or may not be tied to a physical place. Yet, observations of these experiences in the online space, including the surrounding physical spaces, are challenging to attain [20,21]. The ways individuals present themselves online likely differ from their in-person identity or their identity online behind the screen. James and Busher [18] recognize the potential of identifying these crossovers between online and offline spaces in the 'in-between' space. This discusses the deep entanglement of space, time, place, and embodiment in the online/offline world that is individual to the person. As Acton [22] demonstrates, this interconnected view of teaching and learning practices is spatialized with unequal relation dynamics. As such, students along with academic staff are agents in the negotiation of the feel or atmosphere of the learning space and should be acknowledged in its management, as they influence the surrounding experience with social structures and systems and their subsequent connection to it. Multi-methods research approaches to data collection can help address these multiple compounding influences, like privacy, feelings, or comfort [18]. Further, employing methods that go beyond textual data can help explore intangible or unspoken qualities, which are often more widespread in the online space [12]. For instance, the recent study by Cook et al. [23] used facial stimuli to demonstrate that zoom backgrounds alter how someone is perceived in terms of their trustworthiness and competence, two factors that influence social connection. Using methods that engage beyond textual data could help understand how student experiences with the learning environment are curated or guided in a way that supports or weakens social connection, as supported by Sumartojo and Pink [24].

This research follows the principles outlined by Wheele et al. [2], who identify four dimensions of affect to help recognize how students build or maintain a social connection at university, namely temporality, adjustability, materiality, and cultural/social norms. Affective atmospheres offer a means of approaching intangible qualities to help explore how student social connection is built and maintained in a digitally oriented university. With the introduction of technology, affective atmospheres help to recognize the possibility of intangible and 'in-between' aspects emerging, like placelessness or being without place, while still influencing an individual's experience of place in this crossover between the online and offline space [13,25]. This can be a collective experience or co-experience, as identified by Batterbee and Koskinen [26], and one that is highly personal. This is because the ability to be moved or altered by an atmosphere is based on both the surrounding space and an individual's affectual history [27]. Thus, using affective atmospheres, the first aim of the article is to address research question RQ1, namely "How is student social connectedness built and maintained in spaces of the digitally-orientated university?".

1.1.2. Step 2: Curating Student Social Connection

The problem of curating student social connection is addressed in the second aim of this article by addressing RQ2, namely "How could a framework support the management of student social connection in digitally-orientated university learning environments?". Academic staff and students have been negotiating these digitally orientated university spaces largely without guidance in terms of how they function or are experienced. This is not ideal, as identified in research like [5], which recognizes the benefit of all stakeholders in the university system being given adequate training to boost digital competence, such as 'netiquette' sessions. However, this has been challenging to implement within university management and faculty development, since there is a general lack of guidance on how to manage and curate the qualities that alter student experiences of social connection for a positive university experience. This makes it difficult for university management to support lecturers and program leaders when guiding students to build and maintain social connections as part of the university experience and campus development. Del Vecchio et al. [28] acknowledge the potential of technology in combining online with offline

experiences through the term 'phygital', which could be applied in fields like consulting, strategy, analytics, and experience design. This considers the potential of combining the online and offline environment to create an experience for the user that integrates optimal qualities of online and offline space. This potential to imagine online and offline spaces is explored in the design field of interaction design. Interaction design considers how a user interacts with a product, system, or service in the physical environment. It offers a means of understanding how to transport experiences as feelings, situations, conditions, and stories through products, systems, or services [3]. This identifies a 'why, what, how' principle to design technology-mediated experiences, mainly focusing on people–technology relations [3,29], which could be useful to apply in a framework for curating student experiences with social connection at university.

Exploring how to design for the experience of the user translates into entire ecosystems of space and transcends place-based research [30], where placemaking theories offer similar intentions as interaction design, with a focus on creating positive people experiences. Placemaking offers a three-step principle, namely inspiration, ideation, and implementation [4], which is much like the 'why, what, how' principle of interaction design. In looking at the two theories of placemaking and interaction design together, both offer a three-step principle for curating an experience. The main difference is that interaction design has mainly focused on people–technology relations, while placemaking has mainly focused on people–place relations. Since student social connection in the digitally oriented learning environment requires an understanding of this amalgamation of relations between people–place–technology, uniting these theories can help support its design and management for university campus development. This article explores interaction design and placemaking as a means of managing and curating experiences in the hybrid-style university that supports social connection. Affective atmospheres are used within this as a concept to recognize the qualities that emerge as students build or maintain social connections. Merging these perspectives from interaction design and placemaking offers a means of framework development, where experiences in the digitally oriented university are managed and curated to support social connection. Framework development is the second aim of this article, using an innovative multi-method approach to develop the research and explore RQ2, namely "How could a framework support the management of student social connection in digitally-orientated university learning environments?".

2. Theoretical Framing: Interaction Design and Placemaking

The theoretical approaches adopted in this article expand across interaction design and placemaking to address people–place–technology relations. Interaction design and placemaking are separate theories that are used interconnectedly for their theoretical framings of place design (placemaking) and experience design (interaction design). The concept of affective atmospheres is introduced to unite the theories in the way of thinking of people–place–technology relations as interconnected. With the movement towards digitally oriented education, an interaction design understanding of people–technology relations and a placemaking understanding of people–place relations could be useful to apply in the management of the digitally oriented learning environment for student experiences with social connection. Experience is understood in this way as the integration of perception, action, motivation, and cognition [3]. Interaction design offers a means of understanding how to transport experiences as feelings, situations, conditions, and stories through products, systems, or services [3]. Although more often considered through user experience (UX), which has less recognition of the spatial aspect, particularly across the virtual and physical world [31], interaction design has the scope to go beyond products, systems, or services. As Kaptelinin and Bannon [30] recognize, the scope of interaction design encompasses spaces of experience, which recognize the amalgamation of products, systems, and services in space, being both online and offline space. In this, Batterbee and Koskinen [26] recognize the role of people–people interactions through something they call 'co-experience'. Further, the work by Hassenzahl [3] demonstrates the power

of connections between people and technology through the physical and digital space, which acknowledges the role of the environment in a wake-up experience created by bird sounds and sunlight radiating from a digital alarm clock. Although not widely applied in university contexts, Kaptelinen and Bannon [30] recognize the value of interaction design in the learning environment. They recognize that the creation and design of the learning environment are dependent upon the user's choices (such as lecturers) and differ according to aspects such as purpose, social context, and technologies employed. In using this research with placemaking, which has traditionally positioned a greater focus on people–place relations and overlooked the role of technology, this article explores the management of people–place–technology experiences.

Affective atmospheres can be utilized as a concept within interaction design and placemaking to help access the intangible qualities that emerge 'in-between' spaces and alter experience [32]. Although other concepts are also relevant and apply to parts of the study, including the sense of place, the focus on affective atmospheres aims to bring a clearer understanding of more dynamic and emerging qualities. While a sense of place recognizes the multifaceted relationships between people and their environments, the concept of affective atmospheres helps to capture the more fleeting and momentary experiences that are not defined as clearly within a sense of place [19]. Although principally emerging from social science disciplines, the study of effect with the topic of atmospheres has grown over the last two decades and applies across contexts [24]. An atmosphere is a human-centered phenomenon that emerges only with someone being there to experience it [25]. As Anderson [27] states, atmospheres are always in the process of becoming and rely on the affective capacities of the surrounding bodies. This can be a collective experience and one that is highly personal, since the ability to be moved or altered by an atmosphere is based on both the surrounding space and an individual's affectual history. Thus, affective atmospheres emerge as the composition of both the material environment and the affective capacity of bodies. Affective atmospheres can help explore people–place–technology relations and the experience for people by acknowledging the less physically tangible qualities in these spaces [2]. This link between experience and the affective capacity of people is recognized in interaction design, with a user-centered design stimulating a shift towards recognizing users' affective experiences [29]. Within placemaking and planning disciplines, atmospheres can be imagined, constructed, and conceived through affect to curate specific experiences [32]. Despite not being widely explored in the learning environment, Ehret and Hollett [33] recognize the value of appreciating the influence of affect when considering, for example, how learning programs are sustained or not over time. Affective atmospheres as a concept, underpinned in the theoretical framing of interaction design and placemaking, is applied in this empirical study to explore how student experiences with social connection might emerge in this process.

Dimensions of affect offer a means of defining the process of creating an atmosphere [25], as used by Wheele et al. [2] to help recognize qualities that alter how students build or maintain social connections at university through temporality, adjustability, materiality, and cultural/social norms. With placemaking and interaction design, these dimensions are proposed to emerge in a three-step process of experience. The 'why' or inspiration phase consists of qualities that can be adjusted to support the flexibility and personalization that students seek through digitally oriented learning environments. The 'what' or ideation phase identifies the significance of cultural/social norms and values, which need to be aligned to support the student experience with social connection and adapt towards a digitally oriented culture. The 'how', or implementation phase, positions materiality and temporality as key dimensions that help shift the design and management of the space towards social connection, granted that the steps beforehand have been addressed. This is not a linear process, and Figure 1 demonstrates how each part of the network is continually moving and needs to be addressed to ensure the outcome emerges. It is produced to acknowledge these ideas as a three-step process of design that combines aspects of interaction design with placemaking theories through affect to help manage

student social connection at university. This theoretical framework is applied in this empirical study to explore how student experiences with social connection might emerge in this process, which can be comprehended and utilized by university staff, including lecturers, program leaders, and university management, to better support student experiences with social connection.

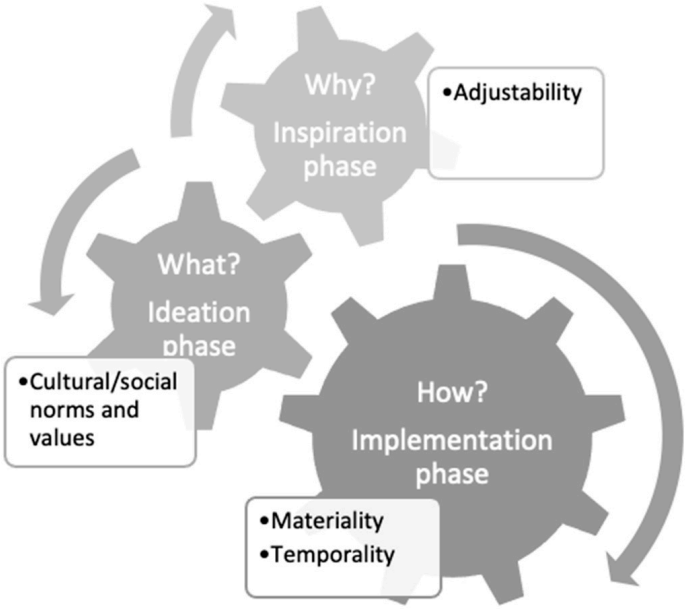


Figure 1. Three-step process of design for managing student social connection at university.

3. Methodology

3.1. Study Design

The study adopts a multi-method qualitative approach, using observations, interviews, and auto-photography. The participants include master-level students (total $n = 68$) and lecturers ($n = 4$) of three hybrid-style courses at two universities in Switzerland and Norway. Classes are coded and identified as Class A (Swiss) shown in Figure 2, Class B (Norway) shown in Figure 3, and Class C (Swiss) shown in Figure 4. Data were collected in stages, with data collected in the autumn semester of 2022 (Class A) and the autumn semester of 2023 (Class B and Class C). Class A and Class C were from the same course but one year apart (Class A in 2022 and Class C in 2023). The main author who conducted data collection remained a silent participant in Classes A and B, although Class C required that they had a semi-active role in taking three student group feedback sessions. Methods of data collection were chosen to address the subjectivities of social connectedness and gain input from the student and lecturer user level in the university system, building on the work of Wheele et al. [2] and supporting the co-design of a guide for supporting student social connectedness in the hybrid-style university learning environment. This produced visual, textual, and audio data, which were analyzed using the thematic analysis method. All forms of data collection were conducted with a varied degree of contribution to answering both two RQs. However, observational data and auto-photography mainly targeted the answering of RQ1, namely "How is student social connectedness built and maintained in spaces of the digitally-orientated university?", while interviews primarily contributed to answering RQ2, which was "How could a framework support the management of student social connection in digitally-orientated university learning environments?".



Figure 2. Physical classroom Class A and Class C in Switzerland (authors own, 2023).



Figure 3. Physical classroom Class B in Norway (authors own, 2023).



Figure 4. Physical classroom Class A and Class C in Switzerland in group table arrangement (authors own, 2023).

3.2. Data Collection

Observational data were collected over the duration of the three courses (approximately five months for each course). This includes over 50 h of observations for each class, totaling approximately 150 h of observations, including lectures, presentations, and group work. Observations were documented through field notes, photographs, and audio recordings. Photographs that include participants are excluded from this article to maintain participant anonymity. There were inherent difficulties in gathering observational data of students during online lectures. While an aspect of feelings behind the screen cannot be accounted for, observations of the online space were made to include qualities like background and number of the video camera, highlighting qualities that students do not necessarily recognize or acknowledge when verbally relaying how connected they feel to the university, their colleagues, or lecturer, e.g., a busy background might make students feel less inclined to speak.

Auto-photography involved asking students to email the main author photographs of their online learning space with a short summary explaining what they liked/disliked about the space and why they used it for doing their university work. All students from each class were asked for their involvement, with a total of n = 17 students choosing to participate and provide a photo. This form of convenience sample brought inherent limitations, where there could be a selection bias toward students who are more open to sharing their learning space. With limited access to the observational data of students beyond the screen during online lectures, photos of students' online learning environment offered insight into the dynamics that occurred behind the screen. This utilized the potential of imagery as a form of data collection, which is a powerful way of exploring qualities that are perhaps overlooked but engrained in and with the student experience online. In reference to Pink [34], visual ethnography offers a means of exploring qualities of the online learning environment that cannot be observed or extrapolated through interview/focus-group data collection. As a critical methodology, ethnography offers a "process of collecting and representing knowledge that is based on ethnographic own experiences" [34], (p. 3). This type of sensory approach is appropriate for the research topic, as demonstrated by Sumartojo and Pink [24], to address the experiences of an atmosphere through relations between people, place, and technology.

Interview sessions were conducted with 4 lecturers from the observed classes. Three lecturers shared teaching in Class A and Class C, and one lecturer solely taught Class B. Interviews were developed with an ethnographic approach, where a semi-structured interview schedule was developed involving 8–10 questions. A collaborative visualization technique was used to help elicit in-depth responses from the participants and build rapport, which helped provoke responses and facilitate the interview [35]. Interviews were documented through audio recordings.

3.3. Field Setting: Course Structure and Classroom

The study was conducted across two universities in Switzerland and Norway, with one being an applied sciences university with a practice-oriented and science-based focus and the other a research university with a technology and science-based focus. The universities were chosen specifically for their provision of hybrid-style courses that fit within the hybrid-style framework. The courses followed were situated in the discipline of real estate as mandatory components of master programs. To ensure consistency, courses were situated within similar fields surrounding the discipline of real estate.

Class A and Class C had a mix of lecturers throughout the course, including guest lecturers. Class B had one lecturer for the whole course. For all classes, in-person lectures took place in the same location each time with a flexible seating arrangement. All classes had a traditional class layout with tables and chairs in rows or groups, but Classes A and C occasionally changed the layout to include high desks and chairs or desks (see Figures 2 and 4). The configuration used was typically taken over from previous classes, although lecturers could request a certain style in advance. Online classes were conducted

in all classes using the same online videoconferencing platform (Communication Platform A) chosen by the lecturer, which students either took alone or in groups in their homes, at the university, or in their workplaces. Different open-source learning management systems were used for the courses (Communication Platform B), which were chosen by the university. This meant that Class A and Class C in Switzerland had the same system and Class B in Norway had a similar but different system. The students typically conducted online sessions either at home, at the university, or at their workplace. Students also used an additional unofficial online communication platform to have informal/private chats (Communication Platform C), which were chosen by the students. These systems varied among students and classes. Communication platforms are coded to avoid any conflicts of interest.

3.4. Participants and Recruitment

A convenience sample was used to recruit students in the master courses chosen, and lecturers based on their involvement with the master courses. Recruitment continued until data saturation was reached. The participants in the observations include students from 2 different courses (n = 68), and the participants in the interviews include the lecturers from those 2 different courses, including guest lecturers (n = 4). This mix of user levels was crucial to integrating the perspectives of different populations in the university system. A mix of languages was spoken in the courses, with English, Norwegian, and German used. Geographically, most students lived locally to their university (within 60 min to access), although some lived further afield (over 60 min to access).

3.5. Data Analysis

Observational data and interview data are analyzed using an orthographic method. The participants remain anonymous in the process to ensure the study remains within the ethics approval agreed to with the NSD (Norwegian Research Council). A semi-deductive approach was applied to the analysis that combined a data-driven inductive approach with a deductive a priori template of themes approach. This is suited for this qualitative type of analysis, which is based on the conceptual stance of the 3-step placemaking framework, which is being developed through a data-driven inductive approach for student social connectedness in the hybrid-style learning environment. The auto-photography tasks produced visual data in conjunction with the accommodating written summary provided by participants.

The underlying theoretical perspective and epistemological stance are based on place-making from within an architectural and human geography background. It is further positioned within a broader epistemological framework of critical realism, which reflects the challenges of trying to gain knowledge on an objective reality, where data might not necessarily bring the answers [24,36]. This acknowledges the innately social nature of knowledge as something that is socially constructed.

4. Finding and Curating Student Social Connection

4.1. Adjustability: Inspiration 'Why' Phase

Although students were made aware that documenting photos of their online learning space did not require composition beyond simply showing the learning space and its situation in the room, students often used the opportunity to clear their desks for the photo, "I must admit that I have cleaned up my room a bit, but it actually looks more or less like this every day". Some mentioned that it gave them the opportunity or excuse to clean their desk. This was not the purpose or point of the study, but it indicates a level of representation and guardedness, with individuals controlling what they wanted to show and what they wanted to keep private. As indicated by Wheele et al. [12], the private space is an influential space when building and maintaining social connections. This private space is hard to access in the online space, as students have greater power to adjust the accessibility to it. Trying to build connections through online space was indicated as difficult for lecturers

without engagement and getting to know the student in person first, making it hard for the lecturer to help support the student, "sometimes I get something just out of the blue and I kind of just think yeah okay who is this person what are they writing on" (lecturer, Class B). Whereas, when a student continually made their presence shown in classes both online and in person, it was indicated to be easier for the lecturer to help support them. This level of familiarity grew during the semester and was largely dictated by the students' capacity to adjust the environment towards their needs, "some other person that's active and has the camera on and all that stuff it's totally it feeds on itself yeah in the way and it's much easier to take to give them some extra help" (lecturer, Class B).

In onsite classes, these interactions between students and the lecturer emerged more frequently, with students choosing seats in the classroom that faced the lecturer or choosing to raise questions with the lecturer during breaks. These interactions also transferred to student-to-student interaction and student-to-place interaction, with break times used as informal spaces 'in-between' to adjust the environment or speak with others in the room. For instance, windows were typically kept closed during onsite lectures, but students would often go to open them during breaks, demonstrating a breakdown of formality or rules as experienced during the lecture. Further, students would often move closer to a plug socket or adjust their desks towards the lecturer during a break time rather than during the lecture. The informal space that emerged in the onsite break times allowed students to build or maintain connections with others or the surrounding environment more freely. These adjustments were not as visible in the online space, being more private or not as socially oriented. During breaks online, students nearly always turned their cameras off and did not interact through the main communication video conferencing platform. Headphones were often seen in photos of students' online learning spaces, suggesting that students used this as a means of controlling the sound around them, enabling students to dictate how they listen or speak even if they do not have silence around them. In the photos, two students shared their online learning space. But, the majority used a quiet place alone to conduct their online learning, as one student mentioned privacy was important, "I live in a shared flat and therefore don't have a separate study. I don't have the distance to my private life (also spatial separation). As a result, I am often distracted and sometimes can't concentrate fully on university" (student, Class B). Ensuring that private spaces exist both online and onsite where students feel comfortable and open to interact was necessary, where one lecturer mentioned that "those that have been active in both the physical and the virtual domain have gotten more out of it" (lecturer, Class C), suggesting that engagement is not only necessary in person but also online.

4.2. Cultural/Social Norms: Ideation 'What' Phase

Structures or rules were found to vary between hybrid-style classes, students, and lecturers. These structures or rules did not always seem well understood, with most lecturers and students improvising as they went along, stating "it's a mix of them testing out stuff and having discussions just playing around with technology" (lecturer, Class C) and "so far when I had online sessions, I didn't have a real strategy to involve students since actually I'm gathering experiences myself" (lecturer, Class B). This lack of clear structure was constantly being negotiated and renegotiated as part of a process, "we even made a communal decision that we should always have the video on but still not all people have it on" (lecturer, Class C). In one class, a lecturer tried to organize online group work presentations with the whole class, but students decided to only participate for their presentation, missing the opportunity to support their colleagues or learn from them, "I'm not sure if they misunderstood it or if they just went okay so we'll just show up when we're presenting" (lecturer, Class B). With these given rules or social structures in online and offline spaces being broken, it suggests that underlying forces encouraged the restructuring of rules and structures, making it acceptable to break, and almost expected since very few students frequently used the video camera. It was most frequently the norm to not use the video camera than to use it. Lecturers mentioned that they were trying to learn how to

engage students without force but in a guiding way, "how can I kind of not force them to be there but to motivate them to be there" (lecturer, Class A). There appeared to be different perceptions of what was expected and how much students were required to participate, with students having some sense of freedom to decide this themselves.

Experiences with the video camera also varied among students in the different classes. In the classes in Norway, students often joined online together, compared to in Switzerland, where students only joined classes online alone and typically used a private space with no other people around to physically take the online class. Whereas students in Norway often physically took the online class in a public area with people walking behind the camera or with others from the course (typically up to four students together). There were also differences in how the students presented themselves with the video camera on, with students in Norway seeming to be more relaxed by not having a strict dress code (e.g., wearing relaxed clothing), eating in front of the camera, walking around with the camera, and leaving the camera and joining again without interrupting the class/announcing it. This initiative was also recognized and appreciated by the lecturer when the students negotiated group feedback sessions, with some students just joining online without asking the lecturer, "they didn't ask me is it OK to bring him in [online], of course it is, just here he is, that's a great thing" (lecturer, Class B). The spontaneity and freedom of being online ensured that students could dictate their own rules and decide how to participate. This could also be owing to the lecturer, who encouraged the students to feel comfortable through the video camera by not forcing a certain image on the students, "we also had people that were kind of almost wandering around downtown, when I asked if they would turn on the camera they were just sitting outside sunbathing and that's totally fine as well" (lecturer, Class B).

4.3. Materiality: Implementing 'How' Phase

In the onsite lecture rooms, all classrooms had views that were unique to the physical classroom, including a lake view and a view of the inner halls and the lecturer's private rooms. This 'personality' of the space in relation to its physical place was unique to both the university and the location of the classroom within the university. It brought a sense of familiarity and shared space between students, lecturers, and their environment. Online, this sense of shared space was lost. In student photographs of their online space, the students chose to sit in spaces that had expansive views outside a window or filled the space with personal qualities, like photos, their dog, seasonal decorations, or personalized furniture. One student also mentioned that their desk at home was 'made out of the same tree as I used to play on when I was a kid' (student, Class A) (see Figure 5). This indicated an intricate connection between the working space and the user, where one student mentioned that they could "always relax my eyes a bit by looking out of the window" (student, Class C), and several mentioned that it helped to work near the window "because of the lighting conditions" (student, Class A), "fresh air when I need it" (student, Class A), and having "a lot of natural light" (student, Class B). Without a co-experience while accessing the materials in these spaces, the online space became individualized, and it was harder for others to connect without a shared experience of space. The lecturers acknowledged that it was harder to build or maintain social connections if the students did not turn the video camera on. Not only was this because they found it hard to know if students understood the content "it is very hard for me to know whether I'm hitting home" (lecturer, Class C), but they also experienced a loss of interaction with the students and "really missed the feedback" (lecturer, Class A). However, rather than students simply not connecting online, it was found that students had instead created group chats using informal communication platforms that they often used during online lectures to have more private or informal conversations.



Figure 5. Student example of online learning environment (students own, Class A, 2022).

The power of technology as a learning tool was recognized by lecturers, but it was often overlooked as to how technologies were engaged with for building or maintaining connections. Technologies had the power to enable private or informal communication between students and the lecturer during or outside lecture hours. In onsite classes, technology also had the power to dictate the atmosphere of a classroom. In one onsite class, there was a series of misunderstandings and inconsistencies between various students and the lecturer, which led to an uneasy atmosphere, general confusion, or distrust towards the lecturer. The lecturer later described the experience as a "breakdown of social connection" (lecturer, Class C). The atmosphere in the class was uncomfortable and tense, forcing students to express how they felt directly to the lecturer. As the lecturer tried to help the students understand, the projector stopped working. This disruption of technology further exacerbated this divide between the students and their trust in the lecturer. During the break, many students left the room, and it was noted that the classroom was left with "cables all around the floor and bags everywhere, it looked a mess" (researcher observational field notes, Class C). This disheveled physical appearance of the classroom replicated the mental confusion of the class prior to the break. This sense of disconnect was experienced by the class as a collective feeling. Thus, while at the moment it brought division between the lecturer and the students, the students were united by this feeling. For this reason, onsite classes were typically more effective at addressing any misunderstandings because the lecturer could build trust with the students by being physically there, moving around, and addressing students directly to get to the bottom of a problem. These small conflicts were important and represented the building of social connection, where without this sense of unguardedness, these conflicts did not emerge, "if you don't have trust then you can't also have conflict" (lecturer, Class A). This was attempted in online classes, but students often ignored the lecturer and avoided talking, even if they had been directly named and asked to. Addressing how to use the virtual space in a way that supports these more intense interactions that are conducive to building or maintaining connection is more complex. Opening access to more informal and private spaces can easily be blocked by the student, as the online space enables a sense of disembodiment. This also indicates the different materiality and temporality of the online space, being somewhere that students can pick up and leave. It was recognized by one lecturer that students might not always understand when to participate or what is needed and, subsequently, might make bad decisions based on their feelings at that moment, "sometimes they make the correct decision not to show

up for but at some point maybe they also make the wrong decision by not showing up" (lecturer, Class B). This was also reflected upon by one lecturer witnessing a student walk out of a lecture, which felt more hurtful in person than online as "the threshold for jumping in and out is so much smaller" (lecturer, Class C). For these reasons, the lecturers recognized the benefit of students engaging with the video camera, with one lecturer mentioning that "it's much better with the camera from a beach than no camera from I don't know" (lecturer, Class B). By lecturers providing motivation at the beginning of lectures to use the video camera (or beforehand via email), typically more students turned the video camera on and kept it on for longer. Supporting and guiding these cultural/social norms that encourage social connection through technology was important for the class, "with people who turn cameras on you just can see them [. . .] it's not about controlling anything it's just about having a good lecture" (lecturer, Class B). The continual dividedness over the video camera highlights the challenge of building or maintaining connection without the trust that it requires to allow others into more private or informal spaces of the virtual space.

4.4. Temporality: Implementation of 'How' Phase

Lecturers found that teaching online altered the teaching pace, where seeing student responses visually made it easier to move through the material quicker, "I could speed up when I see how they are reacting" (lecturer, Class A). However, even if students had the video camera on, it was hard to see student responses or engage with them. This was especially apparent when all of the students turned the video camera on, and it was impossible to view all of the students. In the learning space onsite, there were significantly more verbal utterances, facial gestures, and hand movements to support and enhance spoken words. Further, body stance and position were often altered onsite in a way that encouraged interaction, such as orientating the body forward and towards the person speaking. This created a certain energy in the space in-between people, which would also be affected by using the laptop to better clarify a point by moving it toward the lecturer or other students. These subtle queues happened at rapid and almost unnoticeable speeds in a way that subtly indicated that an individual either did or did not support or understand the other students or the lecturer. This was not prototypically private, but also not obviously public, and exhibited a sense of liminality in an 'in-between' space where these interactions played a significant role in supporting communication. This contributed to building a connection, but it was recognized by lecturers as a process that takes time to recognize and respond to these cues in a way that supports connection, "when you are teaching or learning and then you have to figure out OK what is this so it takes time" (lecturer, Class B). Online, a large majority of these actions were not only slowed down but lost, as indicated by the lecturers "[on-site] I see their faces, feel the atmosphere in the room, which I do not so much live online and some have their cameras switched off" (lecturer, Class C). For lecturers, figuring out how to act or how to encourage certain behaviors was recognized as something that is processual, shifting, and emerging over time, "some of that was planted in the initial thing and it was nurtured along the way" (lecturer, Class B). In this way, the learning space was found to be dictated in a continual process between students, the lecturer, university managers, and the environment.

In the online space and with the loss or slowing down of many social cues, figuring out what supports these connections requires a different approach. Looking at the online learning space for the 'in-between' informal or private spaces that might support social connection, it was found that student backgrounds provided a glimpse of student personality when the video camera was used. For instance, one student had a bike hung on the wall behind them, which could offer a topic of mutual interest as a starting conversation. These glimpses of the informality of a student's private space were temporary, changeable, and dictated by the student in terms of what is shown. Thus, while aspects of social connection might be supported by certain online backgrounds (both student and lecturer), as discovered in the study by Cook et al. [23], finding how to manage this in the university space is difficult without the acceptance of its users. It was described by

one lecturer that connecting with the students in the lecture is part of a process, where you must begin right from the beginning by sharing what the students' part or role in the lecture requires and how they will contribute to the class and the atmosphere surrounding it. These relationships had to be planted, nurtured, and let grow, "you have to initiate it and then you have to kind of nurture it" (lecturer, Class B). Lecturers recognized that there needed to be time at the beginning of a course dedicated to laying out the "mode of collaboration and have same understanding of the importance of doing different or sticking to different rules" (lecturer, Class C). Further, a guide or course on online engagement would be useful to support the creation of these social structures or rules, "an education for teachers and students together, reflecting about what mode of collaboration interaction is needed to have good teaching and good learning experiences [. . .] in the beginning of the studies" (lecturer, Class C). There were also differences recognized between lecturer connections to students across different cohort years, "last year I remember having talks to students on campus and um sometimes made a joke or try to tell them and if you don't get it let's have a second session" (lecturer, Class C). This was also reflected in the way that students dealt with complaints, where in previous years, they had come to the lecturer directly. This time, they went straight to the program leader, demonstrating a lack of trust in the lecturer. This breakdown of trust could have been encouraged by a reduction of in-person student–lecturer interactions that were out of the lecturer's control, where they later indicated that "you need a minimal amount of meeting face to face to create social connection" (lecturer, Class C). This represents the difficulty for university management to control how the digitally oriented university learning environment can be used to help guide toward social connection without direct access to altering the behaviors in this space.

5. Discussion

The following section addresses both research questions through the findings, "How is student social connectedness built and maintained in spaces of the digitally-orientated university?", and "How could a framework support the management of student social connection in digitally-orientated university learning environments?". First, student social connectedness is found to be built and maintained in both the tangible and intangible spaces of the digitally oriented university learning environment. Tangible and intangible spaces are in between open and closed spaces, with qualities shifting and emerging based on people–place–technology relations. Emerging in the informal and private spaces both online and physically, these spaces have qualities that support students in building and maintaining social connections. The online space is noted as less accessible than the physical spaces. The qualities are outlined in a wheel-of-qualities diagram by using real-world examples taken from the study. Second, student social connection is found to be built and maintained in stages, which are fluid and changeable but dependent on each other in a continual process of regulation. This is explored and demonstrated in a three-step framework of experience, offering a compass for lecturers, program leaders, and key decision makers for guiding toward student social connection at university for campus development.

As demonstrated by students in this study, building or maintaining social connection was typically sought and appreciated, typically in the informal or more private spaces of being in person. This was shown through frequent interactions and amicable greetings between students and the lecturer in class.

5.1. Accessing Spaces of Connection

Interactions and amicable greetings were sought and appreciated between students and the lecturers in informal and in-person physical spaces which demonstrated building and/or maintaining social connection. Such interactions were not witnessed as frequently in the online space, suggesting fewer instances of building and/or maintaining social connections. There is also a possibility that building and/or maintaining a social connection was emerging in the online space but was not accessible to the researcher, indicating that building and maintaining social connection online also occurred in more private or informal

spaces, like in person. For instance, students often acknowledged that they had tidied their online learning space before sharing a photo of it. This demonstrated a desire to control the privacy of the space and ensure it maintained a level of formality that they deemed acceptable for an online learning space. These components that they used to display their ideal image of the space emerged in both tangible and intangible ways. One student highlighted the tie to their childhood with their desk being carved from the wood of a childhood tree, indicating a sense of home. With the materiality of the desk, the sense of home was continually present and emerging, transported through not only the materials of the space but also by the connections to the person. This intimate connection indicates how a place might be intangible and felt but also presented through the tangibility of an object like a desk. Without the student giving the researcher access to this connection, it would remain a private experience of the space. This inaccessibility to connection occurs through technology, but technology also offers the possibility for these connections to be made accessible with the individual's permission/desire.

James and Busher [18] find that there is a deep entanglement of space, time, place, and embodiment in the online/offline world that is individual to the person. This was illustrated in how students use their physical surrounding space (e.g., their home desk) when using the online learning environment to create crossovers of experience to the physical world, where a sense of connection might be hidden and private or outside the formal lecture view. Students have greater flexibility and personalization in the online space than in the physical space. In the online space, rules or social structures were regularly broken or adjusted, such as, for instance, using the video camera. As such, accessing these spaces for social connection was often conducted experimentally by lecturers and students. Students often turned to more private or informal forms of communication online to connect, which excluded the lecturer. These behaviors of using communication technology that turned towards more private and informal spaces for building or maintaining social connections seem to reflect the behaviors demonstrated in onsite lectures, although this emerged less frequently or visibly online. These spaces enabled a sense of privacy and informality to emerge, where students could seek connection online in blurred, liminal, and largely intangible experiences of space. As identified by Burkell et al. [13], technology opens the potential for 'in-between' space to emerge. Thus, while the study by Cook et al. [23] recognizes the role of backgrounds in a video call, managing this in the digitally oriented learning environment is found in the study to be more complex, requiring engagement from all user levels. Addressing UNESCO [10] and their recognition of the need for technology access, governance regulation, and teacher preparation for technology in education to be used to its full potential, there is a challenge that the informal and private space of online space is not fully accessible or governable. Using technology in education requires an understanding of how to build and maintain student connections in the 'in-between' informal and private spaces online that are largely intangible to access.

Turning to cultural/social norms that existed in the 'in-between' informal and private spaces online, many students did not feel comfortable with opening access to their online learning environment, even if this was the expectation or rule. Found also by Burkell et al. [13], exceptions emerge to a general rule that can subsequently cause underlying alterations to the rule. In this study, the lecturer offered support and encouraged students to join online with their video camera, even preferring them to join with the camera 'on a beach' rather than not at all. However, students still avoided putting the camera on or turned it off after the initial request to have the camera on was no longer the main topic of conversation. This is also demonstrated within the data collection of this study, as a limited number of students provided a photo of their online learning space, despite the researcher asking on several occasions both online and in person.

There are implications of the underlying rules and structures between students and their desire to open their experience of social connection through a hybrid communication culture at university. This requires that a culture of communication supports social structures that do not restrict social connection. Considering the opportunities given by materiality and temporality in the 'in-between' informal and private spaces online, students used these moments and tools to create group chats using informal communication platforms. Similarly, the use of private and informal places in person was used to build and/or maintain connection. In these spaces online that tended to support social connections that included students, it was much harder for lecturers to see or have any influence over the space, and this excluded the lecturer. For instance, onsite students could be more subtle about their engagement, such as choosing seats in the classroom that faced the lecturer or raising questions with the lecturer during breaks. Thus, managing this 'in-between' space online to support rather than weaken student social connection required lecturers to constantly 'check-in' with students, which often felt forced, and students did not engage. Online, the students could more easily hide behind the protection of the screen, where there was no real threat or tangible consequence if they avoided engagement in the formal space. This highlights how technology acted to disembody individuals, providing greater intangibility, where there were no real consequences of disengaging or avoiding connection. This disengagement is similarly identified by Burkell et al. [13], who found that information shared in public online spaces did not involve sensitive or truly personal concerns. With this disembodying of people in space, the lecturer needed to apply greater encouragement to ensure they could still build a connection with students. This was important because student–lecturer connections needed to be built and maintained over time to enable them to accurately judge whether a student was understanding, build a level of trust, and ensure the student became recognizable, making it easier to help them in the long term. This disconnect between the 'in-between' spaces of informal and private space online in comparison to the physical in-person space altered how student social connection was experienced.

Figure 6 uses examples from the study to demonstrate a wheel of qualities influencing students' building and/or maintaining social connection in the digitally oriented university. This aims to identify student social connection, bringing light to the 'how' implementation phase of the 'who, what, how' principle and the three-step process of placemaking framework. The inner ring represents the three indicators of social connection (socializing, social support, and a sense of belonging), which are used in the study to identify/seek social connection. Addressing the spatial role of the digitally oriented learning environment, the middle ring represents the 'how' implementation phase, with intangible qualities in the 'in-between' spaces, which are harder to detect or measure through the three indicators of social connection (socializing, social support, and a sense of belonging). These indicators are recurrent in the outer ring of the wheel (the 'how' implementation phase), which pinpoints how certain qualities could become more tangible and influence aspects of social connection in the 'in-between' spaces. Surrounded by a rotation of the four dimensions of affect, this wheel offers a representation of the multiplicity of possibilities that can emerge in a processual experience of people–place–technology interactions. This offers examples of how these tangible and intangible qualities emerge as part of an interrelated process, which is sought and identified by the dimensions of affect and the three indicators of social connection. This wheel offers real-world examples to campus planners and facility management services of the possibilities of student experience with social connection within the digitally oriented university.

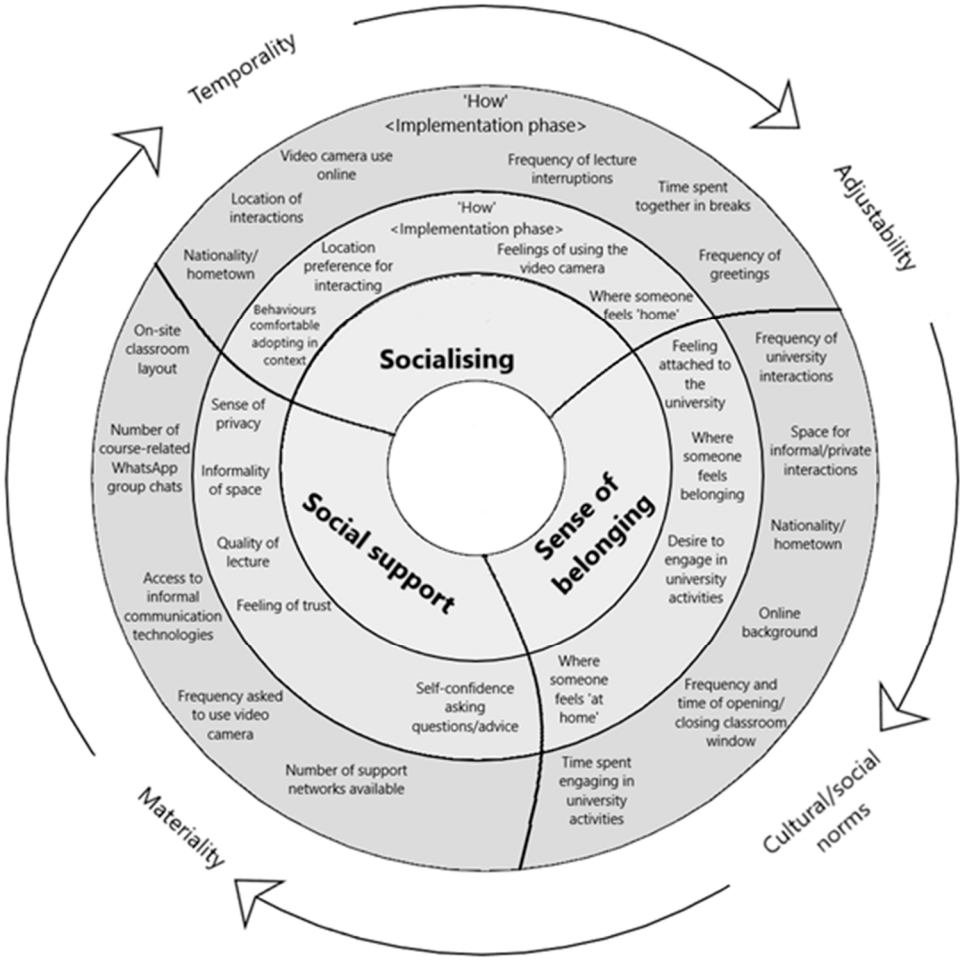


Figure 6. Wheel of qualities influencing students building and maintaining social connection in the digitally oriented university.

5.2. Framework of Experience

Dimensions of affect offer a means of identifying and defining the process of creating an atmosphere and an experience for student social connection, as used by Michels [25] and later adapted by Wheele et al. [2] through adjustability, cultural/social norms, materiality, and temporality. The findings from this study are used to demonstrate how these dimensions of affect can be applied in the theoretical framings of placemaking and interaction design to curate the experience, with qualities emerging within each stage of the dimensions. With placemaking and interaction design, these dimensions are proposed to emerge in a three-step process of experience, which is demonstrated in Figure 7.

First, the 'why' or inspiration phase demonstrates an ability to be adjustable in its nature. This phase consists of qualities that can be adjusted to support the flexibility and personalization that students seek through digitally oriented learning environments. While the inspiration phase sets the end goal in the placemaking literature, within affective atmospheres, the end goal is adjustable and alters based on human experiences. Campus planners can acknowledge the role of this elasticity of hybrid-style space in future development strategies. These align with the digitally oriented university learning environment, being both a private and public space that emerges online and in-person. This brings a sense of liminality, as the spaces become 'in-between', being not quite open and not quite closed. This can shift the nature of formality in the space, with different levels of comfort bringing different experiences with formality or access to formality. This alters how social connections are built and maintained, which is important to address in spatial management and planning in the university learning environment.

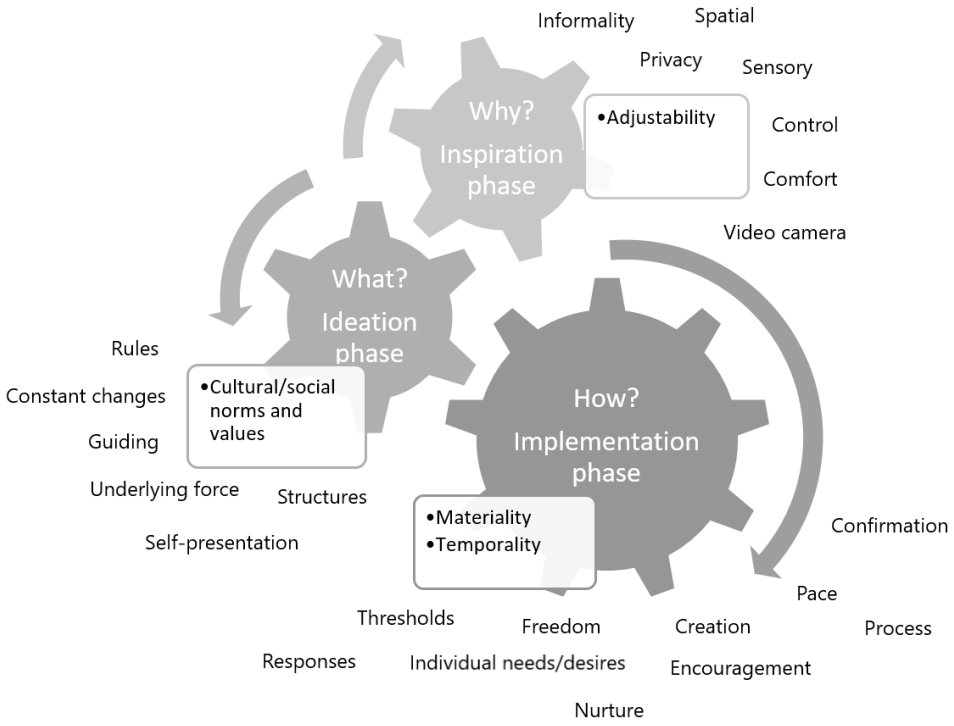


Figure 7. Three-step framework of experience for student social connection at university.

Second, the 'what' or ideation phase constitutes the social qualities that turn a space into a place. This identifies the significance of cultural/social norms and values, which need to be aligned to support the student experience with social connection and adapt towards a digitally oriented culture. Addressing these cultural/social qualities in the learning environment as integral to university planning and management processes would help recognize the role of individual and collective experiences in campus development. This ideation indicates that spaces for social connection come from external influences of support, whereas a sense of place is more internal to individual ideations of the space based on their feelings. The implementation phase emphasizes intangible and tangible qualities that can be influenced by thick places in terms of individual meaning and thin places in terms of space affordability. These recognize the temporality of space as being shaped in terms of time and materiality through underlying forces and guiding structures.

Third, the 'how' or implementation phase positions materiality and temporality as key dimensions that help shift the design and management of the space towards social connection, granted that the steps before are also being addressed as part of the process. This is not a linear process, and Figure 1 demonstrates how each part of the network is continually moving and needs to be addressed within campus development strategies to ensure the outcome emerges. Figure 7 is produced to acknowledge these ideas as a three-step process of experience that combines aspects of interaction design with placemaking theories through affect to help manage student social connection at university. This theoretical framework is applied in this empirical study to explore how student experiences with social connection might emerge in this process. Within campus planning and management strategies/services, this can be comprehended and utilized by university staff, including lecturers, program leaders, and university management, to better support student experiences with social connection.

Addressing research on the topic of combining technology in online and offline experiences, such as that by Del Vecchio et al. [28], this research offers the potential to apply in practice to university campus development. This has impacts across multiple areas, including spatial planning and management, real estate, facility management and services, campus planning, and experience design. It does this by addressing the potential of

combining the online and offline environment to create an experience for the user that is supported, which is important in the learning environment where social connection is a key part of that experience for students. By addressing research on this topic and developing the three-step framework of experience, the value of this work intends to emerge in the learning environment both in theory and practice for university campus development.

6. Limitations and Conclusions

The three case studies offer a relatively wide perspective of different student populations and individuals across different countries. The specific focus on individual perspectives was the purpose of the study to enable deep and personal understandings, but it would benefit from wider research on different student cohorts outside the disciplines of real estate or also different study levels with different teaching/learning structures, both online and offline. During data collection, it was not possible to attain student engagement with all stages, and this was a drawback to the study. However, it also demonstrated the challenge of accessing these intangible spaces in which social connection is built and maintained in the digitally oriented learning environment. Since not all students participated in all stages, the convenience-sample approach subsequently brought limitations, where there is a risk of selection bias towards students who are more open to sharing their experiences with social connection. Although this is a limitation, it also offers findings and opens the opportunity to develop future studies and develop the research.

In this article, the key focus is to understand how student social connection is built and maintained in a digitally oriented university structure and develop a framework that can be used like a compass to guide lecturers, program leaders, and university management to better support student experiences with social connection. The article demonstrates the importance of understanding the spaces 'in-between' offline and online learning spaces, which often consist of intangible qualities that are hard to access, as they often emerge in the informal or private spaces for students building and/or maintaining social connection. These informal/private spaces were challenging to recognize or find online, where it seemed that students had a sense of control over what they revealed and what they kept private when building or maintaining social connections. For instance, there was a general avoidance and dislike of being seen through the video camera, despite students recognizing that the video camera also brought them social connection. These conflicting feelings demonstrate the problems surrounding communication technology, privacy, and connection in the online space, where control and accessibility are heavily granted by the user. This sense of control over what is revealed and what is kept private was also replicated in the actions that emerged in the offline space, but communication technologies made it harder for lecturers, program leaders, and university management to access and extend opportunities for social connection. This adjustability of the digitally oriented learning environment made it more flexible and personal for students, which is deemed to be largely the appeal of this style of learning. Yet communication cultures in the online space have not kept pace with the changes moving toward learning between online and offline, and the underlying rules and structures have encouraged students away from using communication technologies to build or maintain social connections. Acknowledging the materiality and temporality of these 'in-between' informal/private spaces opens the opportunity for using technologies that support informal/private spaces of interaction online and within a time process that encourages, rather than restricts, social connection.

Author Contributions: Conceptualization, T.W., C.L., T.H., C.W. and L.W.; methodology, T.W.; validation, T.W., C.L., T.H., C.W. and L.W.; formal analysis, T.W.; investigation, T.W.; data curation, T.W.; writing—original draft preparation, T.W.; writing—review and editing, T.W. and C.L.; visualization, T.W.; supervision, C.L., T.H., C.W. and L.W.; project administration, T.W. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: The study was conducted in accordance with the Norwegian Research Council (NSD) and approved by the Institutional Review Board (or Ethics Committee) of the NSD (report reference number: 457427 valid until 1 June 2025).

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The datasets presented in this article are not readily available due to privacy restrictions. Requests to access the datasets should be directed to therespw@stud.ntnu.no.

Acknowledgments: We would like to thank the Campus for the Future project at NTNU and the ZHAW Institute of Facility Management for kindly supporting and funding this research.

Conflicts of Interest: The authors declare no conflicts of interest.

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